Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	0.4 [Ω] (1) Contact resistance / resistance of connecting wires and or crocodile clips (1) Don't accept internal resistance / natural resistance		1	1	2	1	2
		(ii)	**Any 3 × (1) from:** • Straight line • Positive gradient / as length increases, resistance increases • Points close to line of best fit / little scatter in the points • Positive intercept so doesn't agree / if 0.4 subtracted line would go through origin so agreement • Graph neither confirms nor denies dependency on ρ or A but is consistent with constancy of $\frac{\rho}{A}$ Award no marks for directly proportional			3	3		3
		(iii)	[Digital] callipers / micrometer (1) Accept screw gauge 0.01 mm **unit mark** (1)	2			2		2
		(iv)	Gradient determined to be 11.6 ± 0.1 or 0.116 ± 0.001 (1) Gradient = $\frac{\rho}{A}$ (can be implied) (1) Area = $\pi(0.115 \times 10^{-3})^2 = 4.2 \times 10^{-8}$ [m²] (1) $\rho = 4.9 \times 10^{-7}\ \Omega\ \text{m}$ **ecf** on gradient and area (1) **unit mark** -1 mark for arithmetical slips of power of 10 / factor of 2 slip **Alternative for first and second marking points:** Use of point from line to give R value with 0.4 Ω subtracted and corresponding l value (1) Substituted into $\rho = \frac{RA}{l}$ (1)			4	4	4	4

Question			Marking details	Marks available					
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	(b)		Avoids heating [the wire] / small current (1) So resistance does not change / does not increase / is not affected (1) Accept reverse argument. Don’t accept any reference to burning or fire risk			2	2		2
			Question 1 total	**2**	**1**	**10**	**13**	**5**	**13**